

4 Safety Engineering Quiz ANS

1. In the deck's terminology, which term describes a hazard with malicious or adversarial intent?

- A) Threat
- B) Failure mode
- C) Exposure
- D) Vulnerability

ANS: A) Threat

2. According to the two-factor risk equation, risk is computed as:

- A) the probability of a hazard multiplied by the severity of its impact
- B) the severity of an accident minus the ability of the system to cope
- C) the sum of a hazard's exposure and the vulnerability of those exposed
- D) system reliability divided by the number of components in the system

ANS: A) the probability of a hazard multiplied by the severity of its impact

3. The detailed disaster risk equation expands the classic risk equation into which four factors?

- A) Probability, severity, redundancy, and transparency
- B) Hazard, threat, failure mode, and fail-safe
- C) Reliability, nines, tail, and drift
- D) Probability, severity, exposure, and vulnerability

ANS: D) Probability, severity, exposure, and vulnerability

4. In the slippery-floor example, which factor increases when more people walk across the wet floor and it stays wet for longer?

- A) Ability to cope
- B) Exposure
- C) Severity
- D) Vulnerability

ANS: B) Exposure

5. A system has 99.9% reliability. How many nines of reliability does it have, and roughly how many decisions are expected before a mistake?

- A) 1 nine; about 10 decisions
- B) 3 nines; about 1000 decisions
- C) 4 nines; about 10,000 decisions
- D) 2 nines; about 100 decisions

ANS: B) 3 nines; about 1000 decisions

6. By what factor does expected time before failure change when reliability improves from 90% to 99%?

- A) It roughly doubles
- B) It stays about the same
- C) It increases about tenfold
- D) It increases by about 9 percent

ANS: C) It increases about tenfold

7. Which example from the deck illustrates the safe design principle of redundancy?

- A) Plugging an AI system into the internet with high-level administrator access
- B) Shivering when the body is too cold
- C) One technician in charge of all three lab chemicals that explode when combined
- D) Vehicles that contain multiple braking systems

ANS: D) Vehicles that contain multiple braking systems

8. Which safe design principle says that each agent should have only the minimum power needed to complete its tasks?

- A) Separation of duties
- B) Principle of least privilege
- C) Defense in depth
- D) Negative feedback mechanisms

ANS: B) Principle of least privilege

9. Which AI fail-safe does the deck suggest?

- A) A confidence estimate that blocks decisions below a set confidence threshold
- B) Removing human oversight so that no delays can occur
- C) A policy of giving the AI as much information as possible so it never hesitates
- D) A single general agent that controls every process from start to finish

ANS: A) A confidence estimate that blocks decisions below a set confidence threshold

10. Which of the following illustrates antifragility?

- A) Lifting weights strains muscles, so the muscles grow stronger
- B) Seeking the opinions of multiple medical experts
- C) Saving important data on multiple hard drives
- D) A thermostat that switches a heater off when a room gets too warm

ANS: A) Lifting weights strains muscles, so the muscles grow stronger

11. What risk does the deck note about using many layers in defense in depth?

- A) Additional layers always make the whole system less reliable than a single one
- B) Layers cannot be combined with redundancy within the same system
- C) Each added layer removes the need for monitoring the individual layers
- D) The layers may interact in unpredicted or undesired ways

ANS: D) The layers may interact in unpredicted or undesired ways

12. According to the Swiss Cheese Model, safety can be improved by:

- A) Reducing the size and number of holes and increasing the number of slices
- B) Aligning the holes in every slice so that hazards pass through quickly
- C) Enlarging the holes so that failures are easier to notice and fix
- D) Removing slices so that the overall system stays as simple as possible

ANS: A) Reducing the size and number of holes and increasing the number of slices

13. How does Fault Tree Analysis proceed?

- A) Start from a single component and predict every outcome it could eventually cause
- B) Average the failure rates of every component to estimate overall system safety
- C) Start from a specific negative outcome and work backward to its possible causes
- D) Assume accidents happen only when one component fails entirely on its own

ANS: C) Start from a specific negative outcome and work backward to its possible causes

14. In the excerpts from James Reason, why are latent failures prime candidates for principled risk management?

- A) They can be removed completely through sanctions and exhortations aimed at individuals
- B) They exist long before an accident, created by upstream organizational decisions
- C) They disappear once equipment has been fully automated by the organization
- D) They occur only at the sharp end, among front-line operators on the day of the accident

ANS: B) They exist long before an accident, created by upstream organizational decisions

15. What is a tail event?

- A) An outcome close to the average
- B) An event that is always anticipated by experts
- C) A high-probability event with small consequences
- D) A low-probability but high-impact occurrence

ANS: D) A low-probability but high-impact occurrence

16. How does the deck define a black swan?

- A) Any event whose estimated probability is below one percent in a given year
- B) A well-understood and frequently observed risk that organizations deliberately accept
- C) A tail event that is largely unanticipated, because it is an unknown unknown
- D) A frequent event whose timing and size can be fully predicted from past data

ANS: C) A tail event that is largely unanticipated, because it is an unknown unknown

17. Which description fits "unknown unknowns" in the known/unknown matrix?

- A) Things we are aware of and understand, such as facts and requirements
- B) Things we are aware of but do not fully understand, such as open questions
- C) Things we do not understand and are not aware that we do not know
- D) Things we do not realize we know, such as tacit knowledge

ANS: C) Things we do not understand and are not aware that we do not know

18. What does the turkey problem illustrate?

- A) A long record of past safety is strong evidence that the future will also be safe
- B) Gathering more observations eventually removes any possibility of a surprise event
- C) Risk estimates become more accurate as the number of past observations grows large
- D) A worldview that explains all past evidence can still be overturned by a black swan

ANS: D) A worldview that explains all past evidence can still be overturned by a black swan

19. In the lottery example (99.9% chance of winning \$15 and 0.1% chance of losing \$10,000, versus 99.7% and 0.3%), what lesson does the deck draw?

- A) A higher chance of winning always means a positive expected value, however large the possible loss
- B) A small change in probabilities can flip the expected outcome, so cost-benefit analysis is unreliable without accurate estimates
- C) Probabilities matter little when the stakes of the possible outcomes are very large
- D) Expected-value calculations are most reliable when the underlying probabilities are quite uncertain

ANS: B) A small change in probabilities can flip the expected outcome, so cost-benefit analysis is unreliable without accurate estimates

20. Why does the deck call "delay our decision to gather more information" a fallacy for long-tailed risks?

- A) More information always makes a decision worse than acting early
- B) Long-tailed risks are easy to estimate without gathering further data
- C) Waiting for more information can mean waiting until it is too late
- D) Delay is always cheaper than acting on a risk that might not materialize

ANS: C) Waiting for more information can mean waiting until it is too late